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Project Title

Community Detection–Based Analysis of Trust and Fraud Patterns in a Social Network

Abstract

Online trust networks play a crucial role in platforms where users interact, rate, or transact with one another. However, such networks are often vulnerable to fraudulent or malicious behavior, which may be difficult to detect using traditional analysis techniques. In this project, we propose a graph mining–based approach to analyze trust and fraud patterns in a social network by leveraging community detection methods. Using a real-world trust network dataset, we model users as nodes and trust relationships as edges. We apply the Louvain community detection algorithm to uncover latent community structures and analyze how distrustful or suspicious interactions are distributed within and across communities. By combining community-level analysis with node centrality measures, this project aims to identify structural patterns associated with potentially fraudulent behavior. The results are expected to provide insights into how trust and fraud manifest in network structure and demonstrate the effectiveness of community-based graph analysis for fraud-related investigations.

Problem and Motivation

Trust-based social networks, such as online marketplaces or peer-rating platforms, rely heavily on user interactions to establish credibility. Despite this, malicious users may exploit these systems by behaving fraudulently or spreading distrust. Detecting such behavior is challenging because it often depends not only on individual actions but also on the broader network structure.

Graph mining offers powerful tools to analyze relational data beyond traditional feature-based approaches. In particular, community detection can reveal groups of users that interact more frequently with each other, while centrality measures can highlight influential or structurally important nodes. In trust networks, fraudulent behavior may concentrate within certain communities or emerge at the boundaries between communities.

The motivation of this project is to investigate whether community structure and graph-based features can help explain and highlight trust and fraud patterns in a social network. Understanding these patterns can provide valuable insights for fraud detection systems and enhance the robustness of trust-based platforms.

Objectives

- Model a real-world trust network as a graph and analyze its basic structural properties.
- Detect communities in the network using the Louvain algorithm.
- Analyze the distribution of distrustful or suspicious interactions within and across detected communities.
- Identify potentially suspicious users using graph-based metrics such as centrality and boundary connectivity.

Related Work

Community detection has been widely studied as a fundamental problem in graph mining, with algorithms such as Louvain demonstrating strong performance on large real-world networks. Prior research has shown that fraud and anomalous behavior in networks often correlate with structural patterns, including abnormal connectivity and high betweenness centrality. Studies on trust and transaction networks, such as Bitcoin trust graphs, suggest that malicious users may act as brokers between communities or generate disproportionate negative interactions. This project builds on these ideas by combining community detection with structural analysis to study trust and fraud patterns in a unified framework.

Proposed Methodology

Dataset

We will use a publicly available trust network dataset (e.g., Bitcoin Alpha or Bitcoin OTC), where nodes represent users and edges represent trust or rating relationships. The graph may be directed and weighted, with edge weights indicating the level of trust or distrust. Basic preprocessing will include removing isolated nodes and focusing on the largest connected component.

Techniques and Algorithms

- Graph construction using NetworkX
- Louvain algorithm for community detection
- Centrality measures (degree, betweenness, PageRank)
- Community-level analysis of distrustful edges

Evaluation Plan

The evaluation will be primarily exploratory and analytical. We will assess the quality of detected communities using modularity and analyze community size distributions. We will compare the proportion of distrustful interactions within communities versus across community boundaries. Additionally, nodes will be ranked based on a simple suspicion score derived from centrality and distrust-related features, and the results will be interpreted qualitatively.

Challenges and Resources

Potential challenges include handling noisy or imbalanced trust data and interpreting results without ground-truth fraud labels. These issues will be mitigated by focusing on relative patterns rather than absolute classification. The project will be implemented in Python using NetworkX, pandas, and matplotlib.

References

[1] Blondel, V. D., Guillaume, J. L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. [2] Newman, M. E. J. (2010). Networks: An Introduction. [3] Kumar, S., et al. (2016). Edge weight prediction in weighted signed networks.
